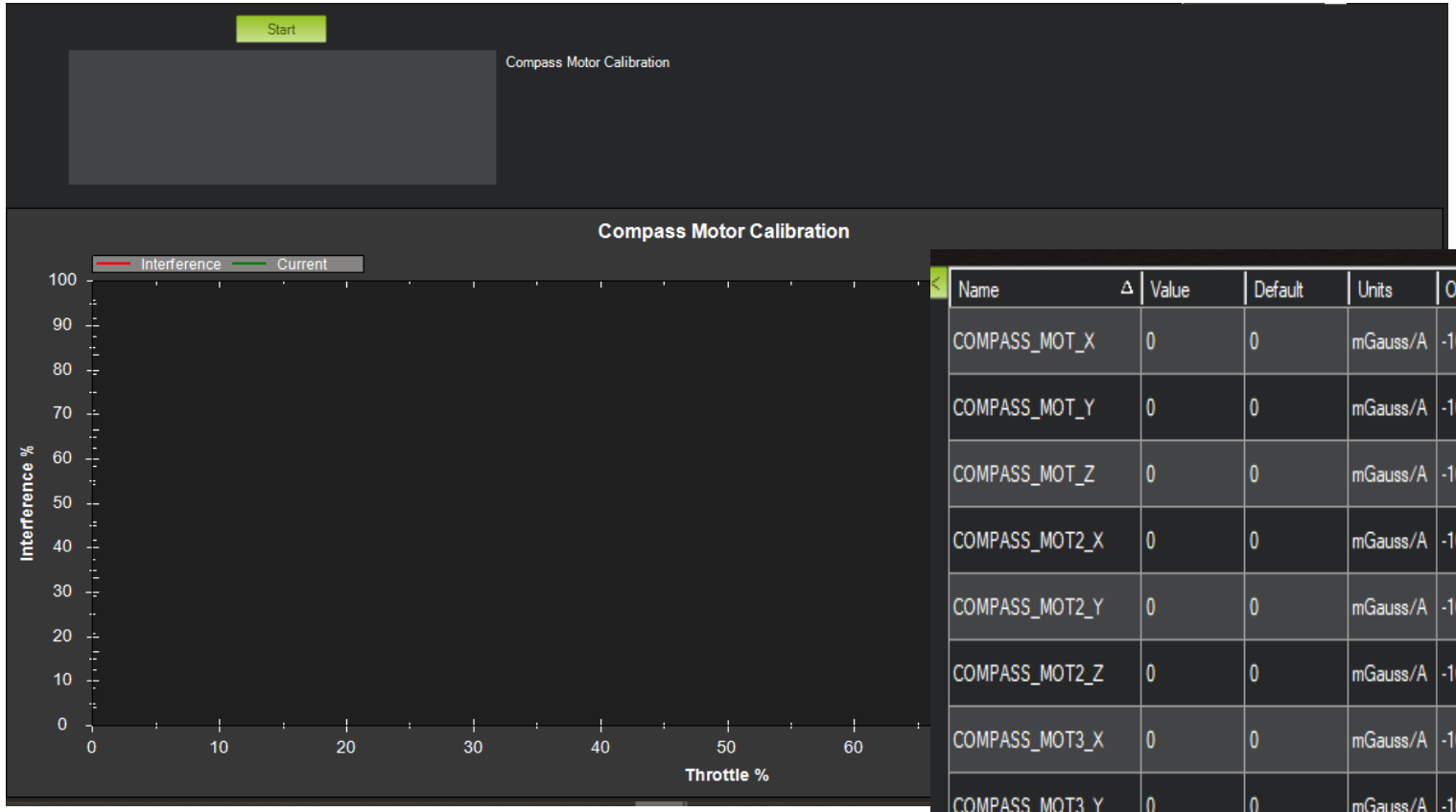
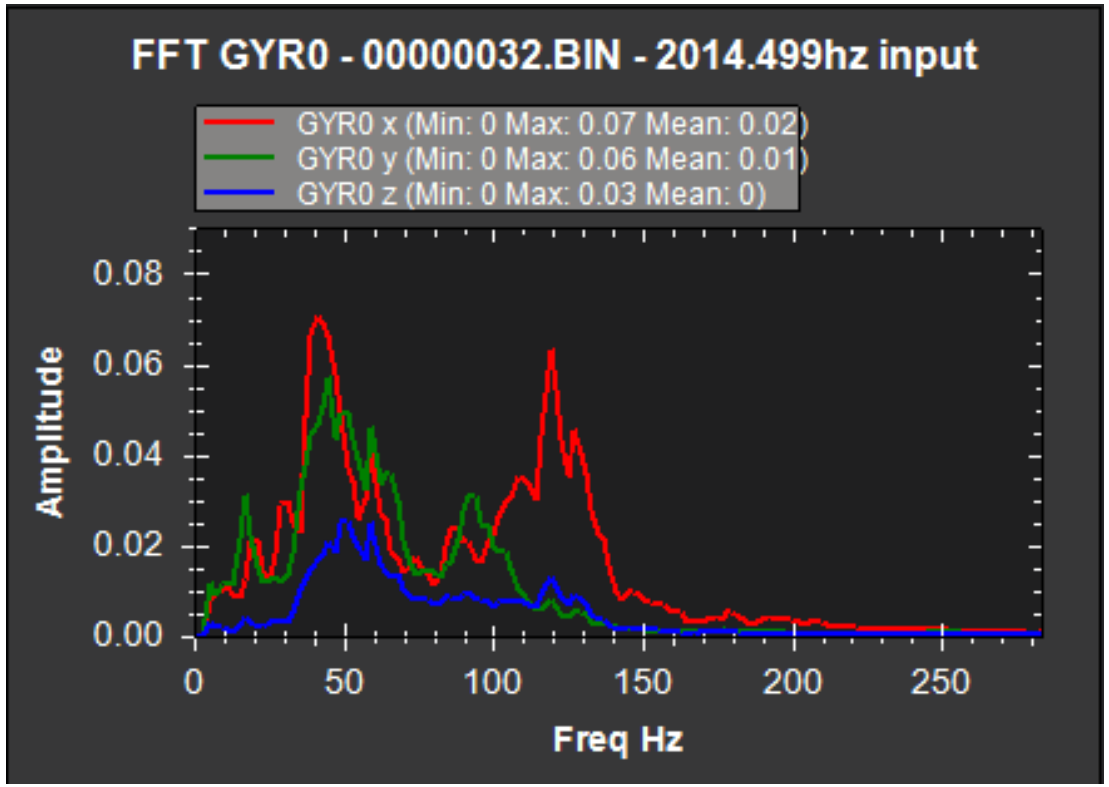
Compass/Motor



Name	Value	Default	Units	Options	Desc	Fav
COMPASS_MOT_X	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to the compass's x-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT_Y	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to the compass's y-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT_Z	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to the compass's z-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT2_X	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to compass2's x-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT2_Y	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to compass2's y-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT2_Z	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to compass2's z-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT3_X	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to compass3's x-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT3_Y	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to compass3's y-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOT3_Z	0	0	mGauss/A	-1000 1000	Multiplied by the current throttle and added to compass3's z-axis values to compensate for motor interference (Offset per Amp or at Full Throttle)	
COMPASS_MOTCT	0	0		0:Disabled 1:Use Throttle	Set motor interference compensation type to disabled, throttle or current. Do not change manually.	

We should see **COMPASS_MOT*** and/or **COMPASS_PMOT*** entries enabled/updated).
Do a short, gentle spin (props still off) and confirm the Compass/Motor page shows reduced interference.

Flight Log Analysis: Tuning the Notche Filter



- **Fundamental:** The tallest, sharp peak is **~47–50 Hz**.
- **2nd harmonic:** Clear peak around **~95–120 Hz**.
- **3rd harmonic:** Noticeable bump around **~140–150 Hz**.
- Beyond 200 Hz, the spectrum flattens, which is typical noise floor.

Flight Log Analysis: Tuning the Notch Filter

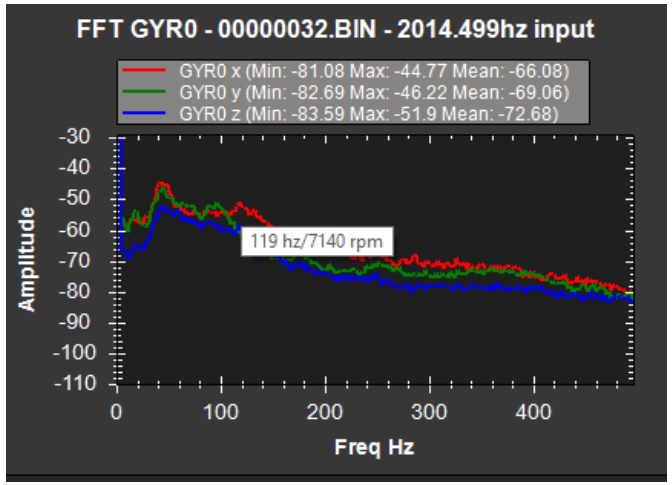
```
# First Notch — unchanged (FFT-based)
INS_HNTCH_ENABLE = 1
INS_HNTCH_MODE   = 4
INS_HNTCH_FREQ   = 47     # seed/min freq
INS_HNTCH_BW     = 23     # half of F1
INS_HNTCH_HMNCS  = 3      # cover 1x, 2x, 3x
INS_HNTCH_ATT    = 40
INS_HNTCH_MASK   = 7
INS_HNTCH_OPTS   = 0

INS_FFT_ENABLE   = 1
INS_FFT_MINHZ    = 30
INS_FFT_MAXHZ    = 200
INS_FFT_OPTIONS  = 1
INS_FFT_SNR_REF  = 10

INS_HNTC2_ENABLE = 0
```

```
# Second Notch — throttle-based, refined

INS_HNTC2_ENABLE   = 1
INS_HNTC2_MODE     = 1  # Throttle tracking
INS_HNTC2_FREQ     = 119
INS_HNTC2_BW       = 30
INS_HNTC2_ATT      = 40
INS_HNTC2_REF      = 0.35
INS_HNTC2_FM_RAT   = 0.7
INS_HNTC2_HMNCS    = 3
```



Flight Log Analysis: ArduPilot Filter Review Tool

INS_GYRO_FILTER 20 Hz SCHED_LOOP_RATE 300 Recalculate filters | Save Parameters | Load Parameters

First Notch Filter

INS_HNTCH_ENABLE 1:Enabled

INS_HNTCH_MODE 4:Dynamic FFT

INS_HNTCH_FREQ 47 Hz

INS_HNTCH_BW 23 Hz

INS_HNTCH_ATT 40 dB

INS_HNTCH_REF 1

INS_HNTCH_FM_RAT 1

INS_HNTCH_HMNCS 3

☑ 1st harmonic ☑ 2nd harmonic ☐ 3rd harmonic
☐ 4th harmonic ☐ 5th harmonic ☐ 6th harmonic
☐ 7th harmonic ☐ 8th harmonic ☐ 9th harmonic
☐ 10th harmonic ☐ 11th harmonic ☐ 12th harmonic
☐ 13th harmonic ☐ 14th harmonic ☐ 15th harmonic
☐ 16th harmonic

INS_HNTCH_OPTS 0

☐ Double notch ☐ Multi-Source ☐ Update at loop rate
☐ EnableOnAllIMUs ☐ Triple notch ☐ Use min freq on RPM source failure

Second Notch Filter

INS_HNTC2_ENABLE 1:Enabled

INS_HNTC2_MODE 1:Throttle

INS_HNTC2_FREQ 119 Hz

INS_HNTC2_BW 60 Hz

INS_HNTC2_ATT 40 dB

INS_HNTC2_REF 0.3499999940

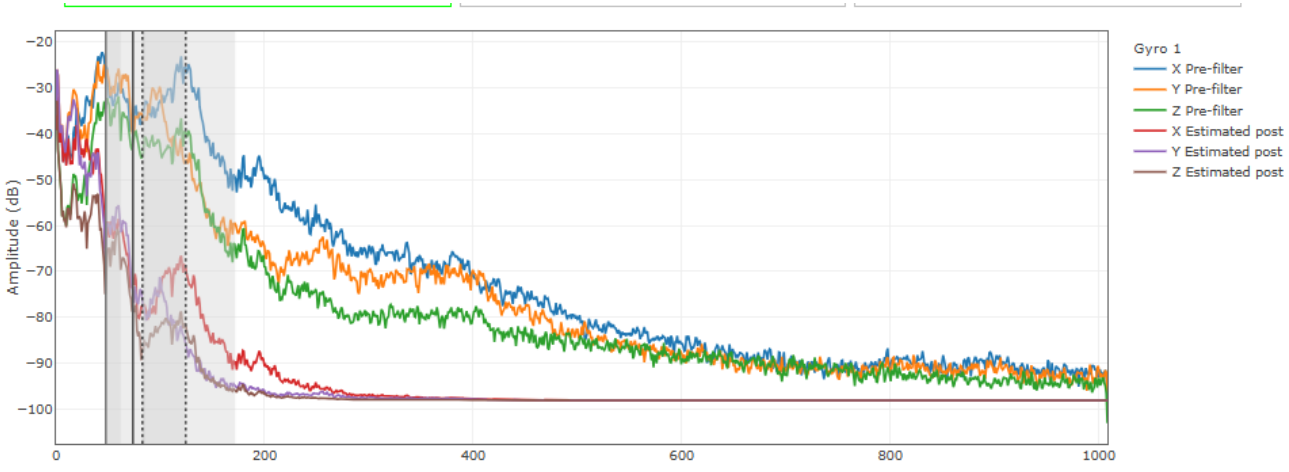
INS_HNTC2_FM_RAT 0.6999999880

INS_HNTC2_HMNCS 3

☑ 1st harmonic ☑ 2nd harmonic ☐ 3rd harmonic
☐ 4th harmonic ☐ 5th harmonic ☐ 6th
☐ 7th harmonic ☐ 8th harmonic ☐ 9th
☐ 10th harmonic ☐ 11th harmonic ☐ 1
☐ 13th harmonic ☐ 14th harmonic ☐ 1
☐ 16th harmonic

INS_HNTC2_OPTS 0

☐ Double notch ☐ Multi-Source ☐ Up
☐ EnableOnAllIMUs ☐ Triple notch



Amplitude scale ⓘ ○ Linear ◉ dB ○ Power Spectral Density

Frequency scale ⓘ ◉ Linear ○ Log ◉ Hz ○ RPM

Notch tracking ⓘ ☑ Show notch 1 ☑ Show notch 2

Loop Rate Aliasing ⓘ ◉ Hide ○ Show ○ Aliasing only

PID Tuning Recommendations (24.5 kg VTOL)

Based

- **RATE vs RATE.Des for Roll (RLL)**
- **RATE vs RATE.Des for Pitch (PIT)**
- **RATE vs RATE.Des for Yaw (YAW)**
- **ATT.DesRoll vs ATT.Roll:** to cross-check the angular controller response
- **VIBE.X, VIBE.Y, VIBE.Z:** to check vibration impact
- **IMU.GyrX/Y/Z** or **NTUN.XKF.FiltX/Y/Z:** (optional) for post-filter gyro clarity

Recommended PID Values (Hover Mode Only)

Axis	Term	Current	Suggested	Comment
Roll	P	0.25	**0.3**	Sharper roll response, better tracking.
	I	0.25	**0.3**	Matches P for steady hold authority.
	D	0.0036	**0.005**	More damping, reduces overshoot.
	FF	0	**0.2**	Reduces lag, improves tracking.
Pitch	P	0.25	**0.3**	Matches roll tuning for symmetry.
	I	0.25	**0.3**	Stronger attitude hold.
	D	0.0036	**0.005**	Adds damping against oscillations.
	FF	0	**0.2**	Sharper pitch command following.
Yaw	P	0.18	**0.25**	Stronger yaw authority.
	I	0.018	**0.05**	Improves heading hold, avoids drift.
	D	0	**0**	Leave at zero for yaw axis.
	FF	0	**0**	Leave at zero for yaw axis.